
DSC5001: Statistical Machine Learning I

Tutorial 4: Linear Regression

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Tutorial Overview

This tutorial practices four aspects of linear regression:

1. Derive the ordinary least-squares solution.
2. Quantify uncertainty in regression coefficients.
3. Diagnose regression problems.
4. Implement linear regression in Python.

Tasks 1–3 are required.

Task 4 is optional.

Task 1: Matrix Form of Linear Regression

Goal. Use the matrix form of OLS to compute coefficients and verify the residual property.
Consider:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

The OLS estimator is:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Tasks:

1. Compute $\hat{\beta}$.
2. Compute fitted values: $\hat{y} = X\hat{\beta}$
3. Verify the residual property: $X^T e = 0$, where: $e = y - \hat{y}$

Solution: Task 1

First compute:

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

and:

$$X^T y = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

Therefore:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

gives:

$$\boxed{\hat{\beta} = \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix}}$$

Solution: Task 1 (continued)

Hence the fitted model is:

$$\hat{y} = \frac{1}{3} + \frac{3}{2}x$$

Using:

$$\hat{y} = X\hat{\beta}$$

we obtain:

$$\hat{y} = \begin{bmatrix} \frac{11}{6} \\ \frac{10}{3} \\ \frac{29}{6} \end{bmatrix}$$

Solution: Task 1 (continued)

The residual vector is:

$$e = y - \hat{y}$$

so:

$$e = \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{bmatrix}$$

Now verify:

$$X^T e = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

OLS residuals are orthogonal to the feature space.

Task 2: Regression Uncertainty

Goal. Understand how the design matrix affects coefficient uncertainty.
Consider a simple regression with:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

Assume: $\sigma^2 = 4$. The covariance of OLS is:

$$\text{Var}(\hat{\beta}|X) = \sigma^2(X^T X)^{-1}$$

Tasks:

1. Compute $(X^T X)^{-1}$.
2. Compute $\text{Var}(\hat{\beta}|X)$.
3. Find the standard error of the slope coefficient.
4. Explain how the uncertainty changes if σ^2 increases.

Solution: Task 2

First compute:

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

The inverse is:

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

Therefore:

$$(X^T X)^{-1} = \begin{bmatrix} 7/3 & -1 \\ -1 & 1/2 \end{bmatrix}$$

Solution: Task 2 (continued)

Using:

$$\text{Var}(\hat{\beta}|X) = \sigma^2(X^T X)^{-1}$$

and:

$$\sigma^2 = 4$$

we obtain:

$$\text{Var}(\hat{\beta}|X) = 4 \begin{bmatrix} 7/3 & -1 \\ -1 & 1/2 \end{bmatrix}$$

Therefore:

$$= \begin{bmatrix} 28/3 & -4 \\ -4 & 2 \end{bmatrix}$$

Solution: Task 2 (continued)

The slope variance is the second diagonal element:

$$Var(\hat{\beta}_1) = 2$$

Hence:

$$SE(\hat{\beta}_1) = \sqrt{2}$$

Solution: Task 2 (continued)

The uncertainty of coefficients depends on:

$$\text{Var}(\hat{\beta}|X) = \sigma^2(X^T X)^{-1}$$

If:

$$\sigma^2 \uparrow$$

the coefficient variance increases.

Therefore:

$$SE(\hat{\beta}_j) \uparrow$$

and confidence intervals become wider.

Coefficient uncertainty depends on both noise level and information contained in the predictors.

Task 3: Regression Diagnostics

Goal. Use residuals, leverage, and Cook's distance to identify influential observations.
Consider:

$$x = (1, 2, 3, 4, 10), y = (2, 3, 5, 4, 15), \text{ and } \hat{y} = (2, 3, 4, 5, 8)$$

Use:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}, \text{ and } D_i \propto e_i^2 \frac{h_{ii}}{(1 - h_{ii})^2}$$

where e_i is the residual and h_{ii} is leverage.

Tasks:

1. Compute residuals.
2. Compute leverage values.
3. Compare Cook's distance values.
4. Identify the influential observation.

Solution: Task 3 – Residuals

Residuals are:

$$e_i = y_i - \hat{y}_i$$

Therefore:

$$e = (0, 0, 1, -1, 7)$$

The largest residual is:

$$i = 5$$

because:

$$|e_5| = 7$$

is much larger than other errors.

Solution: Task 3 – Leverage

For simple regression with an intercept:

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}$$

For:

$$x = (1, 2, 3, 4, 10)$$

we have:

$$\bar{x} = 4$$

and:

$$\sum_j (x_j - \bar{x})^2 = 50$$

Solution: Task 3 – Leverage Calculation

The leverage values are:

$$h_1 = 0.2 + \frac{9}{50} = 0.38$$

$$h_2 = 0.2 + \frac{4}{50} = 0.28$$

$$h_3 = 0.2 + \frac{1}{50} = 0.22$$

$$h_4 = 0.2$$

$$h_5 = 0.2 + \frac{36}{50} = 0.92$$

Therefore:

$h_5 = 0.92$

Solution: Task 3 – Influence

Observation 5 has:

$$|e_5| = 7$$

and:

$$h_5 = 0.80$$

Therefore it has:

large residual + high leverage

which suggests potential influence.

Solution: Task 3 – Cook's Distance

Using:

$$D_i \propto e_i^2 \frac{h_{ii}}{(1 - h_{ii})^2}$$

the residuals and leverage values are:

$$e = (0, 0, 1, -1, 7), \text{ and } h = (0.35, 0.27, 0.22, 0.20, 0.80)$$

Therefore:

$$D \propto (0, 0, 0.008, 0.004, 19.6)$$

The largest value is:

$$i = 5$$

Observation 5 is the most influential point.

Optional Task 4: Regression Modeling and Diagnostics

Goal. Build a regression pipeline and evaluate both prediction performance and model assumptions.

Use the California Housing dataset from scikit-learn.

Tasks:

1. Split the data into training and test sets.
2. Fit:
 - (a) mean-response baselineand:
 - (b) linear regression model
3. Compare test MSE.
4. Plot residuals against fitted values and identify whether any pattern exists.
5. Report the coefficient with the largest absolute value and explain its meaning.

Solution: Task 4 – Data Preparation

Load the dataset:

```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
```

```
data = fetch_california_housing()
```

```
X = data.data
```

```
y = data.target
```

Split training and test data:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2,
    random_state=0
)
```

The test set is kept unseen during training.

Solution: Task 4 – Model Comparison

Mean baseline:

```
baseline = np.mean(y_train)
```

```
y_pred_base = np.ones(  
    len(y_test)  
) * baseline
```

Linear regression:

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
```

```
model.fit(  
    X_train,  
    y_train  
)
```

Solution: Task 4 – Evaluation

Compute test MSE:

```
from sklearn.metrics import mean_squared_error
```

```
mse_base = mean_squared_error(  
    y_test,  
    y_pred_base  
)
```

```
mse_linear = mean_squared_error(  
    y_test,  
    y_pred  
)
```

Solution: Task 4 – Evaluation

Residual diagnostics:

```
residual = y_test - y_pred
```

```
plt.scatter(  
    y_pred,  
    residual  
)
```

```
plt.xlabel("Fitted values")  
plt.ylabel("Residuals")  
plt.show()
```

Solution: Task 4 – Interpretation

Students should discuss:

- ▶ Whether linear regression improves over the mean baseline.
- ▶ Whether residuals show systematic patterns.
- ▶ Which features have large coefficients.

A good analysis should include:

prediction performance + diagnostic evidence

Tutorial 4: Linear Regression

Questions and discussion